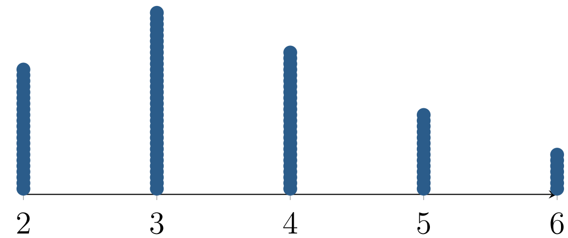
# Can Chance Make a Streak?

Topic 4.1 · TODAY'S PROBLEM

Suppose a coin advertised as fair is flipped 10 times and gives HHHHHHTTTT. To investigate what chance alone could produce, a class simulates 100 sets of 10 independent fair-coin flips. For each set, the class records the longest run of matching outcomes; the dotplot shows the results.



Longest run in one set of 10 fair-coin flips

**Claim A:** "The coin is rigged—six heads in a row cannot be chance."

**Claim B:** "Streaks like this can happen by chance; simulation can tell us how unusual this one is."

**FIRST TAKE (5 min, on your sheet)**

Which claim seems better supported right now? Cite one detail from the sequence or dotplot.

DOK 1 **(a)** State the longest run in the observed sequence and the most common longest-run value among the simulated sets.

DOK 2 **(b)** Use the dotplot to estimate the probability that 10 fair-coin flips have a longest run at least as long as the observed run. Interpret the estimate in context.

DOK 3 **(c)** Which claim is better supported by this evidence? Cite the simulation result that settles your choice. Then describe a larger simulation that would more precisely settle the disagreement: state what one trial is, what to record and count, how many trials to use, and what result would make you doubt that the coin is fair.

**Video + follow-along:** `u4_lesson1-2_live.html` (scan the code)
**Finish (a)–(c) after the video. Turn in your sheet.**